

Gravitational Dynamics

Three-Body Laplace Resonance in Jupiter's Moons

Diogo de Macedo

Stephanie Berntsen Croft



RUC

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Introduction

The combination of Jupiter, its rings, and moons is commonly referred to as Jovian system. From an astronomical standpoint, the Jovian system displays very particular and fascinating features. Jupiter is, by far, the largest planet in the solar system, being about 2.5 times more massive than all other planets of the solar system combined [2]. Another interesting characteristic is that Jupiter is a gas giant, meaning that its mainly composed of gaseous matter, more precisely hydrogen and helium. Recent data recovered from the Juno mission has also shown that the planet is completely covered by clouds and storms [2]. While all these facts are interesting, the main focus of this mini-project will be the planet's four largest satellites, also know as *Galilean moons*: Io, Europa, Ganymede, and Callisto (Figure 1.1). Throughout the theory section their dynamics will be explored, but first let us present their main physical properties. Further details can be found in [6].

- **Io** - Is the innermost and third largest Galilean moon. Unlike the other Galilean moons, Io has a dry surface with no evidence of water presence. The tidal forces exerted by Jupiter and the two nearby moons, Europa and Ganymede, generate heat culminating in extremely intense volcanic activity. In fact, Io is the most volcanically active body in the solar system [7]. To put into perspective, data gathered from pictures and infrared analysis calculated the planets heat flux to be close to 2 w/m^2 , whereas that of the Earth is only around 0.1 w/m^2 [6]. This ultimately results in the planets chemical composition being primarily of heavy metals, displaying silicate rock surrounding a molten iron or iron sulfide core. In terms of atmospheric composition, Io has an very tenuous atmosphere made up mostly of sulfur dioxide [2].
- **Europa** - Is the smallest of the Galilean moons. The surface of Europa is in a frozen state, being covered by a layer of ice, which makes the moon one of the most reflective bodies in the solar system. Data collected by the Galileo spacecraft suggested Europa is, much like the Earth, composed of silicate rock with an iron core and a rocky mantle. However, the existence of tidal forces could potentially cause the ice to melt near Europa's surface, which has led scientists to postulate the existence of an ocean of liquid water beneath the surface, covering a range around 100 km deep [6]. Europa is thought to have twice as much water as does Earth. This moons is particularly relevant for astrobiology studies due to its environmental potential for having micro-habitats such as underwater volcanoes, analog to those found in the Earth, where extremophilic organisms could thrive [2]. Some evidence of cryovolcanism has also been found [2].
- **Ganymede** - Is the largest satellite in the solar system and the only moon known to have its own, internally generated, magnetic field [2]. This magnetic field generated a small magnetosphere insider Jupiter's, larger, magnetosphere. The presence of a heavily crated surface indicates a very active geological past, with some instances of cryovolcanic activity [2]. It also also has an extremely thin atmosphere, mostly composed of oxygen and ozone. The presence of oxygen could be due to ice sublimation on the surface or to the existence of gas bubbles trapped under the ice [2].
- **Callisto** - Is the Galilean moon positioned furthest from Jupiter and the less geologically active. The existence of craters and asteroid crashes on its surface are indicators that the surface has been intact for a long geological time period. This makes sense considering that Callisto has

not suffered the same tidal effects as the other Galilean moons, as a result of being the furthest away from Jupiter and is not in orbital resonance. Its atmospheric composition is theorized to be composed of both carbon dioxide and molecular oxygen [6].

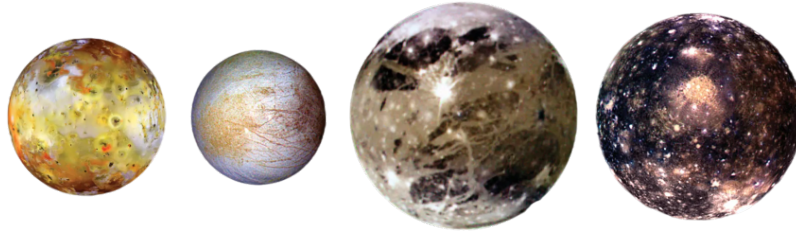


Figure 1.1: The four Galilean moons: Io, Europa, Ganymede, and Callisto, respectively. Image taken from [2].

In general, the orbital mechanics involved in satellite motion is more complex than that of planets. This arises mainly because at small distances some forces become significant and can no longer be neglected (i.e tides or oblateness). In order to have a more solid grasp of the system to be studied, it is important to have in mind some select physical and mean orbital parameters of the celestial bodies, as those provided in table 1.2.

	Io	Europa	Ganymede	Callisto
Physical parameters				
Mass (10^{22} kg)	8.932	4.800	14.818	10.759
Mean radius (10^3 km)	1.821	1.561	2.631	2.410
g surface (m/s^2)	1.79	1.314	1.430	1.235
Density (10^3 kg/ m^3)	3.528	3.014	1.942	1.834
Orbital parameters				
Period (days)	1.7691	3.5512	7.1546	16.6890
Mean motion ($^\circ/\text{day}$)	203.49	101.37	50.32	21.57
Semi-major axis (10^5 km)	4.220	6.713	10.706	18.831
Eccentricity	0.0042	0.0095	0.0013	0.0074
Inclination ($^\circ$)	0.04	0.46	0.21	0.20

Figure 1.2: Select physical and mean orbital parameters of the Galilean system. These are the values to be utilized in the simulation section of this report. Data retrieved from [3].

The pivotal feature of the Galilean satellite system dynamics is that it encompasses *orbital resonance*. By definition, orbital resonance refers to the gravitational phenomenon in which two or more orbiting bodies have orbital periods related to each other by a ratio of small integers [5]. It usually occurs as a result of regular, periodic, and mutual gravitational interactions. The orbital resonance Galilean moons exhibit is a 1:2:4 orbital period ratio [5]. This particular orbital period ratio is commonly referred to as *Laplace resonance*, because french physicist Pierre-Simon Laplace was the first to study and describe the system mathematically. For every orbit Io completes, Europa makes exactly two full orbits while Ganymede makes a total of four orbits around Jupiter. Notice that Callisto is not described here, as it is not in orbital resonance with the other Galilean moons. The dissipative effects of Callisto on the system, while being an exciting ongoing topic of astrophysical research [4, 2, 6], is out of the scope of this project and such effects will not be included in the subsequent computer simulations.

The aim of this mini-project is to develop an octave script that simulates the dynamics of the Galilean system and analyze how much each of the satellites eccentricity deviates when the mass of Europa is increased by a given factor .

2

Orbital Mechanics

Before diving into the development of the computational methods used in the simulations, it is convenient to give a quick overview of the theoretical framework which leads to it. One of the core concepts in astrodynamics is a special case of Newton's second law, called *Newton's Law of Universal Gravitation*. Let us start by considering the simple case of a celestial body of mass m orbiting around the Sun at a given distance r , as illustrated in figure 2.1.

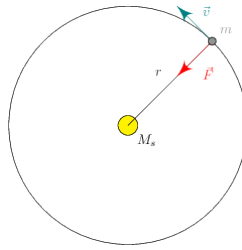


Figure 2.1: A body of mass m in circular orbit at a distance r from the Sun. The centripetal force vector, F , points inwards towards the sun whereas the tangential velocity vector is always tangent to the circular trajectory of the body.

The existence of a centripetal Force, F_c , will cause the body to undergo uniform circular motion and move with constant tangential speed, v , along its orbit. This results in a centripetal acceleration, a_c . The motion of the body is described mathematically by direct application of Newton's second law [1]:

$$\sum F = ma = F_c = ma_c = m \frac{v^2}{r} \quad (2.1)$$

Recall the definition of angular acceleration, ω :

$$\omega = \frac{v}{r} \Rightarrow v = \omega r \quad (2.2)$$

Substituting back in equation (2.1) yields:

$$F_c = m \frac{(\omega r)^2}{r} = m \omega^2 r \quad (2.3)$$

Further writing the angular acceleration as a function of orbital period, T , allows us to express the net centripetal force in terms of both orbital period and orbital radius, as follows:

$$T = \frac{2\pi}{\omega} \Rightarrow \omega = \frac{2\pi}{T} \quad (2.4)$$

Thus,

$$F_c = m \left(\frac{2\pi}{T} \right)^2 r \quad (2.5)$$

Simplifying the equation,

$$F_c = \frac{m4\pi^2 r}{T^2} \quad (2.6)$$

Kepler's third law states that the squared orbital period, T^2 , is equal to the cubed size of the semi-major axis of the planets orbit [1], r^3 . It therefore comes that:

$$T^2 \propto r^3 \Rightarrow T^2 = kr^3 \quad (2.7)$$

Where k is an empirically determined proportionality constant.

Combining equations (2.5) and (2.7) holds that:

$$F_c = \frac{4\pi^2 m}{k r^2} \quad (2.8)$$

Or equivalently,

$$F_c = G \frac{m}{r^2} \quad (2.9)$$

Where the term $4\pi^2/k \equiv G$, and gets coined universal gravitational constant. The net centripetal force acting on the orbiting body is provided by the gravitational attraction exerted by the sun on the body. According to Newton's third law, the body must therefore be exerting a force of equal magnitude and opposite direction on the sun. As shown in equation (2.9), the centripetal force is directly proportional to the mass of the planet, and consequently must also be directly proportional to the mass of the sun, M_S . Hence:

$$\frac{4\pi^2}{k} \propto M_S \Rightarrow \frac{4\pi^2}{k} = GM_S \quad (2.10)$$

Finally, by substituting equation (2.10) back into equation (2.9) Newton's law of universal gravitational is obtained:

$$F_c = G \frac{mM_S}{r^2} \quad (2.11)$$

Where in the general case M_S simply becomes M and represents the value of the mass of the most massive interacting body. Equation (2.11) asserts that the force of attraction between any given set of particles in the universe is varying directly with the product of the masses and inversely with the square of the distance between them.

Another interesting characteristic of orbiting bodies is found when setting equations (2.1) and (2.11) equal to each other, as shown below:

$$F_c = G \frac{mM}{r^2} = \frac{mv^2}{r} \quad (2.12)$$

Solving for the tangential velocity, v :

$$v = \sqrt{\frac{GM}{r}} \quad (2.13)$$

Equation (2.13) describes the constant tangential velocity of an orbiting body and is therefore very useful in astrophysical calculations. It is interesting to point out the rather counter intuitive fact that the mass of the body has no effect on its orbital speed. The orbital speed of any body is merely dependent on the mass of the interacting system and the distance between them.

The most central concept in this project is the so-called orbital resonance. Two, or more, bodies are said to be in orbital resonance if their orbital periods can be expressed as a ratio of integer numbers [5]. This periodic relationship leads interacting systems to exert gravitational influences over each other which ultimately affects the eccentricity of their orbits. This project will focus on the three-body resonance present in three of Jupiter's largest moons: Io, Europa, and Ganymede. Figure 2.2 showcases a visual representation of the Galilean system to be simulated. The orbital resonance this particular system exhibits is a Laplace resonance, characterized by a 1:2:4 orbital period ratio [5]. This means that whenever by the time it takes Io to complete one full revolution around Jupiter, Europa completes two orbits, while Ganymede takes four orbits.

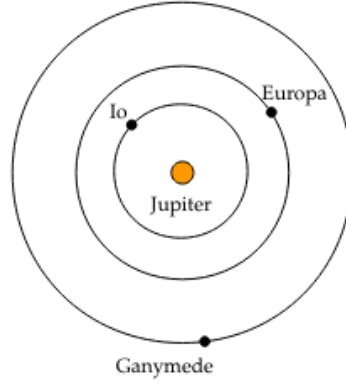


Figure 2.2: The Galilean moons and their positions relative to Jupiter.

Let us now find the equations that describe the dynamics of the moons. The physical quantity that corresponds to the reverse period is named mean orbital motion, n . It is often expressed in degrees per day [4]:

$$n = \frac{2\pi}{T}, \text{ or simply, } n = \frac{1}{T} \quad (2.14)$$

Given that there is a 2:1 ratio in the orbital periods of Io, T_{Io} , and Europa [4, 5], T_{Eu} :

$$\frac{T_{Eu}}{T_{Io}} = 2 \quad (2.15)$$

Expressing the periods in terms of mean orbital motion holds that:

$$\frac{\left(\frac{1}{n_{Eu}}\right)}{\left(\frac{1}{n_{Io}}\right)} = 2 \Rightarrow n_{Io} = 2n_{Eu} \quad (2.16)$$

Subtracting $2n_{Eu}$ to both sides,

$$n_{Io} - 2n_{Eu} = 0 \quad (2.17)$$

Now, substituting the data provided in figure 1.2 into equation (2.17) yields:

$$203.49 - 2 \cdot 101.37 = 0.75^\circ/\text{day} \quad (2.18)$$

The fact that $0.75 \neq 0$ seems to be an indication that the orbital resonance is not perfect, however, that is not the case. The correction on the systems orbit involves the phenomena of perijove precession [4], denoted $\dot{\omega}$. The corrected equations therefore become[4]:

$$n_{Io} - 2n_{Eu} + \dot{\omega}_{Eu} = 180^\circ \quad \text{and} \quad n_{Io} - 2n_{Eu} + \dot{\omega}_{Io} = 0^\circ \quad (2.19)$$

By definition, mean longitude represents the ecliptic longitude at which an orbiting body could be found given that its orbit is circular and free of perturbations [7]. It is expressed in terms of mean motion and perijove precession as $\lambda = \dot{\omega} + n$. Equation (2.19) can then be re-written as:

$$\lambda_{Io} - 2\lambda_{Eu} + \omega_{Io} = 0^\circ = 0\text{rad} \quad (2.20)$$

And,

$$\lambda_{Io} - 2\lambda_{Eu} + \omega_{Eu} = 180^\circ = \pi\text{rad} \quad (2.21)$$

The orbital periods of Europa and Ganymede also follow a 1:2 ratio. Hence, by mathematical analogy it comes that:

$$n_{Eu} - 2n_{Ga} = 0^\circ = 0\text{rad} \quad (2.22)$$

Where n_{Ga} represents the mean orbital motion of Ganymede. Similarly:

$$\lambda_{Eu} - 2\lambda_{Ga} + \dot{\omega}_{Eu} = 0^\circ = 0\text{rad} \quad (2.23)$$

Combining equations (2.21) and (2.23),

$$\phi_L = \lambda_{Io} - 3\lambda_{Eu} + 2\lambda_{Ga} = 180^\circ = \pi\text{rad} \quad (2.24)$$

Equation 2.24 is commonly known as *Laplace relation* [4], and the quantity ϕ_L is defined as *Laplace argument*[7]. Note that equations (2.20) and (2.21) establish the following mean longitude relation:

$$\dot{\omega}_{Io} - \dot{\omega}_{Eu} = 180^\circ = \pi\text{rad} \quad (2.25)$$

A direct implication of equation 2.25 is that the orbits of Io and Europa are anti-aligned, which ultimately makes the occurrence of a triple conjunction impossible [5], as depicted in figure 2.3

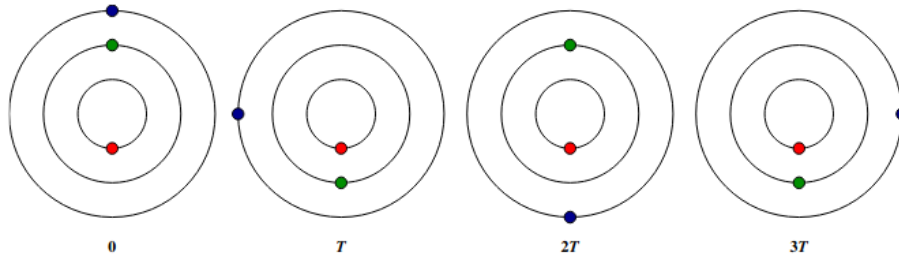


Figure 2.3: Evolution and resonance of the Galilean system, where T corresponds to Io's orbital period. Io is represented in red, Europa in green, and Ganymede in blue. Notice that, as predicted by equation (2.25), Io and Europa's periods are out of phase which prevents all three satellites from ever being in conjunction simultaneously, i.e those satellites cannot lie together in a line on the same side of Jupiter. However, for each Europa-Ganymede conjunction there are three Io-Ganymede conjunctions and two Io-Europa conjunctions [4].

3

Simulation and Analysis

The aim of this report is to simulate how much the eccentricity of Europa varies when setting different values for this satellite's mass. In order to reduce computational overhead, it was decided to scale down the system. This was accomplished by performing a normalization for base quantities of distance, mass, and time. The distance between Jupiter and Io was arbitrarily set to 1 unit, and further normalization was calculated accordingly:

$$\begin{aligned} \text{Io} &= \frac{4.27 \cdot 10^8}{4.27 \cdot 10^8} = 1 \\ \text{Europa} &= \frac{6.71 \cdot 10^8}{4.27 \cdot 10^8} = 1.57 \\ \text{Ganymede} &= \frac{1.07 \cdot 10^9}{4.27 \cdot 10^8} = 2.51 \end{aligned}$$

Similarly, masses were scaled down by normalizing the mass of Io, as follows:

$$\begin{aligned} \text{Io} &= \frac{8.93 \cdot 10^{22}}{8.93 \cdot 10^{22}} = 1 \\ \text{Jupiter} &= \frac{1.90}{8.93 \cdot 10^{22}} = 2.13 \cdot 10^5 \\ \text{Europa} &= \frac{4.80 \cdot 10^{22}}{8.93 \cdot 10^{22}} = 5.37 \cdot 10^{-6} \\ \text{Ganymede} &= \frac{1.48 \cdot 10^{23}}{8.93 \cdot 10^{22}} = 1.66 \cdot 10^{-5} \end{aligned}$$

Additionally, we defined the time scale, T , in terms of the orbital period of Io. Given that Io takes 1.769 days to complete a full revolution around Jupiter, it came that $T = 1.769 \cdot 24 \cdot 3600$.

The first run of simulations is shown in Figure 3.1. There, all parameters were kept constant with the exception of Ganymede's orbital period number. Jupiter is represented as a blue dot with null position and velocity vectors $\vec{x} = (0, 0, 0)$ and $\vec{v} = (0, 0, 0)$, respectively. The orbits of Io (red), Europa (black), and Ganymede (blue) are also shown - note that this color scheme will be consistent throughout the simulation figures. Each of the moons has an initial position vector in agreement with the normalized distances mentioned above. Initial orbital velocities for each moon were calculated according to equation (2.13), taking the form $\vec{v} = \left(\sqrt{\frac{GM_J}{r_m}} \right)$, where G is the gravitational constant, M_J represents the mass of Jupiter, and the distance between Jupiter and the respective moon is given by r_m .

Over the next pages, plots generated by the simulations as well as a subsequent analysis will be given. The full octave code utilized in the simulations of this report can be found in appendix A.

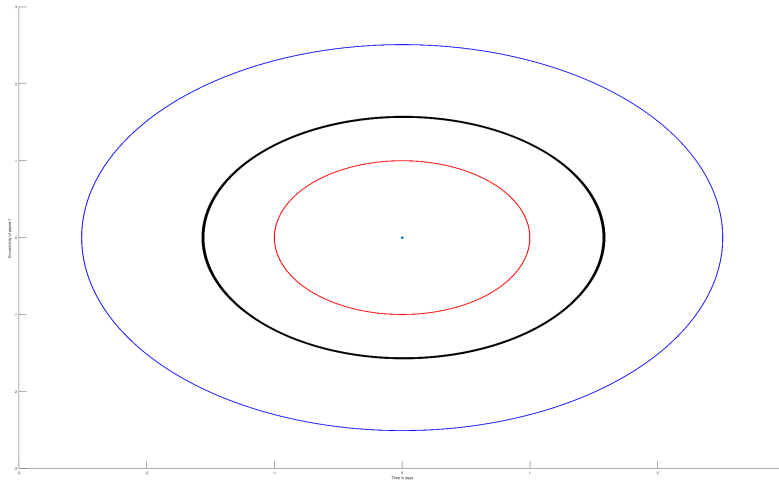


Figure 3.1: Four-body system encompassing Jupiter and orbital trajectories of Io (red), Europa (black), and Ganymede (blue). Each moon's eccentricity is being plotted against time. The run-time of this simulation was of 40 orbital periods of the outer moon, Ganymede. The fact that the starting positions of the moons were along the same line resulted in all the moons displaying elliptical orbits of low eccentricity.

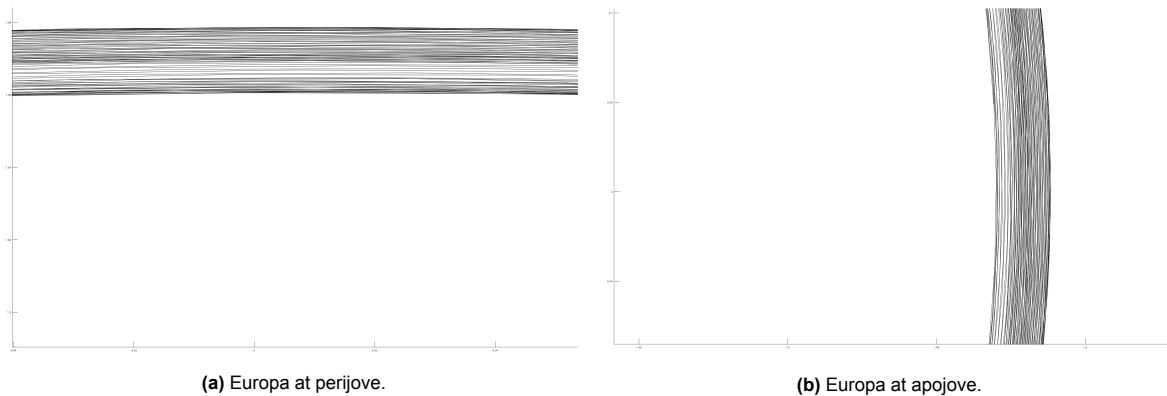
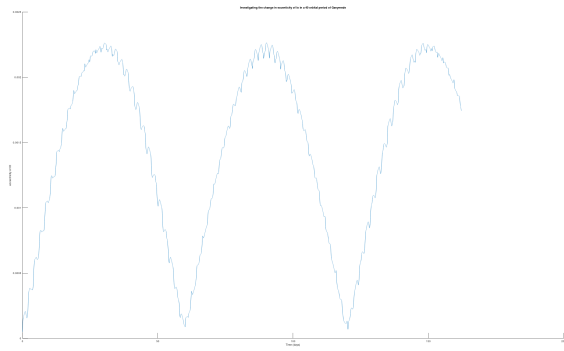
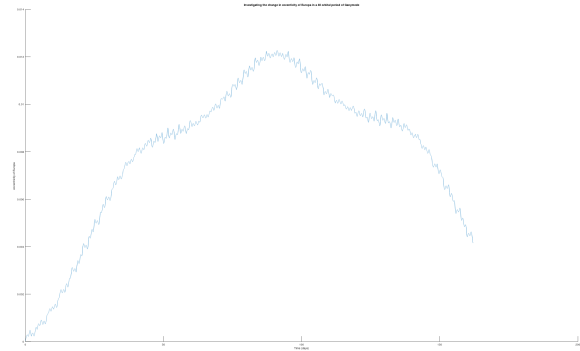


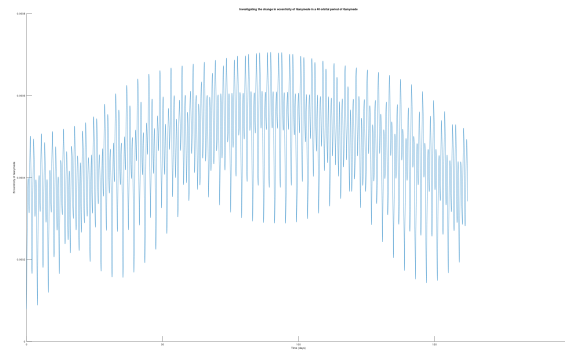
Figure 3.2: Zoom-in of Europa's change in eccentricity around Jupiter through 40 orbital periods of Ganymede. At **(a)** perijove (orbital point closest to Jupiter's center) and at **(b)** apojove (orbital point farthest from Jupiter's center).



(a) Orbital eccentricity of Io versus time. The time for this simulation corresponds to 40 orbital periods of Ganymede.



(b) Europa's eccentricity as a function of time, through 40 orbital periods of Ganymede.



(c) Variations in Ganymede's eccentricity throughout 40 orbital periods.

Figure 3.3: Same simulation parameters as Figure 3.1. Each sub-figure (a)-(c) displays how the eccentricity of the respective moon is affected through 40 orbital periods of Ganymede.

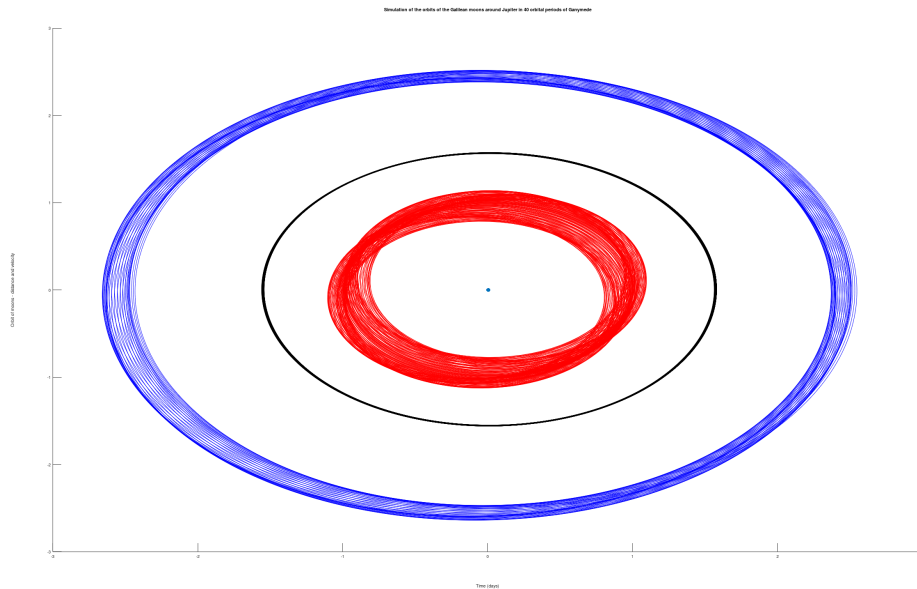
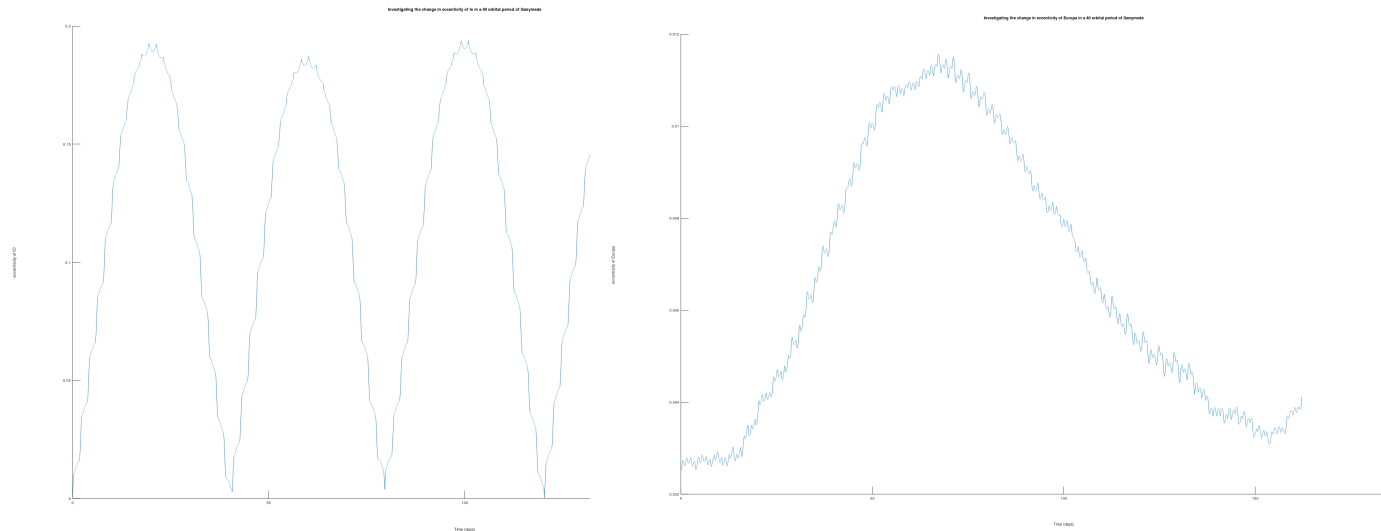
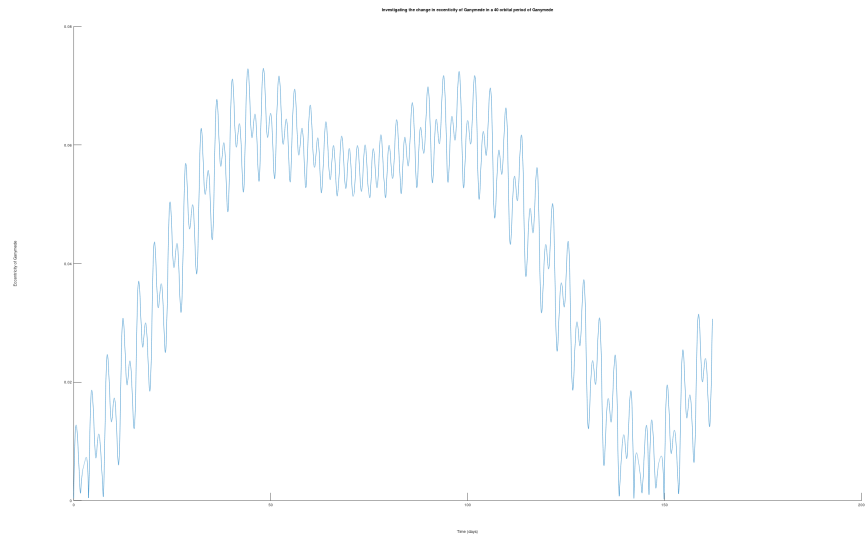


Figure 3.4: Effect of an increase in Europa's mass by an order of magnitude of 2, over a timeframe of 40 orbital periods of Ganymede. Notice a clear shift in orbital eccentricities of both Ganymede and Io as a result of the increased gravitational pull. Europa's orbit, however, seems to remain largely unaffected.



(a) Variance of Io's eccentricity when increasing the mass of Europa by 2 orders of magnitude, over a timeframe of 40 orbital periods of Ganymede. (b) Effect that changing Europa's mass by 2 orders of magnitude has on its own orbital eccentricity. Running time set to 40 orbital periods of Ganymede.



(c) Same simulation parameters as Figure 3.1, where the real value of Europa's mass was considered.

Figure 3.5: Complementary eccentricity versus time plots for the simulations ran with Europa's mass set 2 orders of magnitude higher.

A comparison of figures 3.1 and 3.4 makes it clear that increasing the mass of Europa greatly affects the orbits of both Io and Ganymede. This is to be expected, and is in agreement with the theory covered in section 2. It occurs due to the increased gravitational interaction Europa shares with the outer/inner moons in the Laplace resonance.

With regards to the eccentricity plots, the first observation has been that the time-normalization of the system performed previously is consistent with the simulation results displayed in figure 3.3(c). In reality, it should take Ganymede 288 days to complete 40 orbital periods, as each orbital period corresponds to about 7.2 days. We normalized this by Io's orbital period, close to 1,6 days, which does result in Ganymede taking around 165 days to complete 40 orbits around Jupiter, as the aforementioned figure displays.

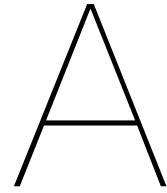
Comparing the changes in the eccentricity of Io in figures 3.3(a) and 3.5(a), the time it took to reach the first peak decreased and the peak got smoother when increasing the mass of Europa. Figures 3.3(b) and 3.5(b) indicate that an increasing Europa's mass by two orders of magnitude decreases its orbital eccentricity by almost 0.01. The shape of the graph also changed, it seems as though in figure 3.5(b) the eccentricity of Europa was kept almost constant until close to the first 20 orbital periods of

Ganymede. The eccentricity of Ganymede, shown in figure 3.3(c), equilibrates around 0.004-0.006, but when Europas mass is increased the system was not able to find an equilibria within the simulated orbital periods.

We believe the initial goal of this mini-project has been achieved. We were able to program a script that accurately simulates the Laplace resonance present in the Galilean system and outputs plots of eccentricity over time for each satellite. While we have developed a suitable model for the desired simulations, we kept the analysis of the system mostly qualitative. This was in part due to the short timeframe to work on the project (two weeks), but mainly a result of the long time it takes to run each of the simulations. However, we deem it important to mention future perspectives and interesting investigations that can be made based on the developed script, provided in the appendix. In order to further analyze the underlying Laplace dynamics of the system, as a first step one could re-run the simulations while altering the mass of Europa more times or with higher orders of magnitude. Ideally, to obtain more meaningful data, the simulation time could also be increased, e.g taking 100 orbital periods of Ganymede. The next step towards a quantitative analysis should be to use mean statistical regression methods on the eccentricity plots of each moons and ultimately reach a model capable of predicting cause-effect of satellite mass alterations in the Laplace resonance of the Galilean system.

References

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Octave Script

The Octave script we developed for this simulations is given below.

```
1  ""
2
3  % This is the Octave script we have been creating to simulate planetary orbits
4
5  G_SI = 6.67408e-11; % Newton's gravitational constant in SI units (m^3 kg^-1 s^-2)
6
7  % Here we choose units for mass, length, time.
8  % This choice affects the numerical value we use for G
9  M = 8.919e22; %IO mass
10 L = 4.271e8; % distance of Io from Jupiter
11 T = 1.769137786*24*3600; % scale for time - how many seconds in the orbital period of Io
12
13 % Relevant masses
14 m_J = 1.898e27 / M; % mass of jupiter
15 m_Io = 8.9319e22 / M; % mass of Io
16 m_Eu = 4.7998e22 / M; % mass of europa
17 m_Ga = 1.4819e23 / M; % mass of ganymede
18
19 % Day length in terms of our chosen time unit
20 day = 24*3600 / T;
21
22 % Gravitational constant in chosen units
23 G = G_SI*M*T^2/L^3;
24
25 % for solar mass/AU/year get the value 4 pi^2
26
27 % Setting initial positions and velocities.
28 % We must have a position (x,y,z) and a velocity (x,y,z) and a mass for each object.
29 % Lets leave out the z-components (Ez=Sz=0) and do 2-dimensional simulations
30
31 %corrected distances
32 R_J=0;
33 R_Io=1;
34 R_Eu=1.57;
35 R_Ga=2.51;
36
37 %velocities - orbital speed depends on how far you are from Jupiter
38 v_J= 0;
39 v_Io= sqrt(G*m_J/R_Io);
40 v_Eu= sqrt(G*m_J/R_Eu);
41 v_Ga= sqrt(G*m_J/R_Ga);
42
43
44 OnePeriode= (2*pi*R_Ga)/v_Ga;
45 numOfPeriodes= 40;
46 runtime= OnePeriode*numOfPeriodes
47
48 positions = [ 0 0 0; R_Io 0 0; R_Eu 0 0; R_Ga 0 0];
49 velocities = [ 0 0 0; 0 v_Io 0; 0 v_Eu 0; 0 v_Ga 0];
```

```

50 masses = [ m_J; m_Io ; m_Eu ; m_Ga ];
51
52
53 nparticles = size(positions, 1);
54 D = size(positions, 2) ; % number of dimensions
55
56 % set the CM velocity to zero
57 momenta = repmat(masses, 1, D) .* velocities;
58 total_momentum = sum(momenta,1);
59 v_CM = total_momentum / sum(masses);
60 velocities = velocities - v_CM;
61
62 % set the time-step
63 dt = 0.0001; % time step in years. Can use day as time step by writing dt = 1*day;
64 % We you could need to adjust this value
65
66 T = runtime;%0; % total time to simulate %we may want to simulate with different values of T
67 nsteps = T/dt;
68 cutoff = 1e10; % not sure it's a good idea....
69 output_file = fopen('orbits.dat','w');
70 output_file_eBin = fopen('eccentricity.dat','w');
71
72 write_interval = 100; % write only every ten time steps to save time
73 %this parameter can be changed to have more/less output
74
75 tStart = tic; % start timer
76
77 % adjust velocities to get them for a half time step backwards
78 % (Leap-Frog velocities are defined at the half-way points between "real" timesteps).
79 forces = gravity(positions, masses, G, cutoff^2);
80 velocities = velocities - 0.5 * (forces ./ repmat(masses, 1, D) ) * dt;
81
82
83 % main simulation loop starts here
84 for s = 1:nsteps
85     forces = gravity(positions, masses, G, cutoff^2);
86     velocities = velocities + (forces ./ repmat(masses, 1, D) ) * dt;
87     positions = positions + velocities * dt;
88     [a, epsilon] = orbital_params(positions, velocities,masses, G, dt);
89
90     if mod(s, nsteps/10)==0
91         printf('\t%2.0f%%\t%8.3f s\n', 100*s/nsteps, toc(tStart)); % print completed
92         % and time spent
93     end
94     if mod(s, write_interval) == 0
95
96         %d_ES = sqrt(sum((positions(2,:)-positions(1,:)).^2));
97         % choose what to write in the output file
98         % the number of items in the format string must match the
99         % number of items following
100         fprintf(output_file, '%6.4f %7.5g %7.5g %7.5g %7.5g %7.5g %7.5g %7.5g %7.5g %7.5g\n',
101             s*dt, positions(1,1), positions(1,2), positions(1,3),
102             positions(2,1), positions(2,2), positions(2,3), positions(3,1), positions(3,2),
103             positions(3,3), positions(4,1), positions(4,2), positions(4,3));
104         fprintf(output_file_eBin,'%6.4f %7.5g %7.5g %7.5g %7.5g %7.5g %7.5g %7.5g %7.5g\n' ,
105             s*dt, a(1), epsilon(1), a(2), epsilon(2), a(3), epsilon(3), a(4), epsilon(4) );
106
107     end
108 end
109
110 tElapsed = toc(tStart) % record time taken for main loop
111
112 % close the file so it can be read in again
113 fclose(output_file);
114 fclose(output_file_eBin);
115
116 % load data from the file
117 o_data = load('orbits.dat');
118 o_dataEcc= load('eccentricity.dat');

```

```

116 t = o_dataEcc(:,1);
117 e1 = o_dataEcc(:,3);
118 e2 = o_dataEcc(:,5);
119 e3 = o_dataEcc(:,7);
120
121
122
123 t = o_data(:,1); % time values
124 % what the columns are depends on what you chosen to write
125
126 xJ = o_data(:,2); % sun's x-coordinate
127 yJ = o_data(:,3);
128 %zS = o_data(:,4);
129 x1 = o_data(:,5);
130 y1 = o_data(:,6);
131 %z1 7
132 x2 = o_data(:,8);
133 y2 = o_data(:,9);
134 %10
135 x3 = o_data(:,11);
136 y3 = o_data(:,12);
137
138 clf;
139 figure(1)
140 hold on
141 plot(xJ,yJ,'*');
142 plot(x1,y1,'r');
143 plot(x2,y2,'k');
144 plot(x3,y3,'b');
145 xlabel ("Time (days)");
146 ylabel ("Orbit of moons - distance and velocity");
147 title([ Simulation of the orbits of the Galilean moons around Jupiter in 40 orbital periods
of Ganymede ]);
148 %title(['eccentricity, r = ',num2str(R_Io),' AU,T ',num2str(T), 'years,Expected nb periodes:
',num2str(numOfPeriodes) ]);
149 %hold off
150
151 %first moon
152 figure(2)
153 hold on
154 plot(t,e1)
155 xlabel ("Time (days)");
156 ylabel ("eccentricity of Io");
157 title([ Investigating the change in eccentricity of Io in a 40 orbital period of Ganymede ]);
158 %title(['eccentricity, r = ',num2str(R_Io),' AU,T ',num2str(T), 'years,Expected nb periodes:
',num2str(numOfPeriodes) ]);
159
160
161 %middle
162 figure(3)
163 hold on
164 xlabel ("Time (days)");
165 ylabel ("eccentricity of Europa ");
166 title([ Investigating the change in eccentricity of Europa in a 40 orbital period of Ganymede
]);
167 %title(['eccentricity, r = ',num2str(R_Eu),' AU,T ',num2str(T), 'years,Expected nb periodes:
',num2str(numOfPeriodes) ]);
168 plot(t,e2)
169
170 %outer moon
171 figure(4)
172 hold on
173 xlabel ("Time (days)");
174 ylabel ("Eccentricity of Ganymede");
175 title([ Investigating the change in eccentricity of Ganymede in a 40 orbital period of
Ganymede ]);
176 %title(['eccentricity, r = ',num2str(R_Ga),' AU,T ',num2str(T), 'years,Expected nb periodes:
',num2str(numOfPeriodes) ]);
177 plot(t,e3)

```